ConsentChain: PureChain-Powered Verifiable Machine Unlearning and Dynamic Consent for Federated IoT Learning

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***Abstract*—Federated learning over consumer IoT devices must now honour data-subject withdrawal under GDPR, the EU AI Act, and CCPA: consent can be revoked and the trained model must demonstrably forget. Prior work treats consent management, machine unlearning, and on-chain auditability as disjoint threads. We close the *consent–unlearning–verification triangle* with ConsentChain, a protocol binding all three in a single chain-enforced pipeline. Its four components are a Consent Lifecycle State Machine (CLSM)—a six-state automaton realised as soulbound ERC-721 tokens; a Verifiable Unlearning Protocol (VUP) that gates on-chain acceptance on three independent membership-inference attacks; a three-contract Solidity architecture deployed on a gas-free (PureChain, PoA^2) and a gas-based (Polygon Amoy, PoS) chain; and a dual-domain evaluation on N-BaIoT and CASAS federations over three unlearning paradigms plus exact retraining. Per-subject lifecycle cost is $0 on PureChain versus $0.0012 on Amoy. The gate accepts approximate unlearning (MIA < 0.55) yet rejects naive retraining—whose distributional uniqueness inflates MIA—and any attestation lacking a 3-of-5 verifier threshold, proving it faithful to MIA evidence rather than the data controller's claim.**

***Index Terms*—federated learning, machine unlearning, smart contracts, dynamic consent, IoT, GDPR, EU AI Act, membership inference attack**

I. Introduction

Federated learning (FL) trains a shared model across decentralised clients without centralising their data [1], often the only legally viable option in consumer IoT. Beyond *technical* privacy (secure aggregation, leakage bounds), a second, equally binding axis remains under-addressed: *regulatory* privacy. GDPR Article 7 requires consent to be purpose-granular and freely withdrawable, and Article 17 grants erasure—which, for a trained model, means it must *forget* a revoked participant; the EU AI Act [2] (Arts. 15/26) and CCPA [3] §1798.105 impose analogous, auditable duties. Each demands demonstrable, audit-grade evidence, not a promise to comply.

Three research threads each address one corner but not the whole. Consent-management ledgers (MedRec [4]) record consent immutably but cannot gate downstream training; machine-unlearning methods (SISA [5], FedEraser [6]) remove a participant's contribution but leave the trigger and verification off-chain; and blockchain-FL systems [7], [8] log model updates on-chain without consent semantics or forgetting verification. No published work binds the three.

We frame the gap as a missing *edge*, not a missing vertex: consent must remain authoritative over which data enters each round, unlearning (by any method) must be *verifiable* rather than self-asserted, and verification against membership inference [9]–[11] must in turn drive the consent state, closing the loop. Prior work implements the vertices; ConsentChain supplies the edges. Doing so off-chain merely moves trust to the controller; a public smart-contract substrate removes it but raises a cost question—a single mainnet revocation costs dollars, prohibitive at IoT scale—which we answer by running the identical protocol on a public, gas-free, PoA^2 chain at *exactly* $0 per operation. This work makes four contributions: (1) a **Consent Lifecycle State Machine** (CLSM)—six states, nine transitions, four formally-stated safety and liveness properties, instantiated as a soulbound ERC-721 contract; (2) a **Verifiable Unlearning Protocol** (VUP), method-agnostic over the unlearning algorithm, accepting an attestation iff three *independent* membership-inference attacks (shadow [9], LiRA [10], label-only [11]) all fall below τ under t-of-n ECDSA signatures; (3) a **three-contract Solidity architecture** (ConsentRegistry, UnlearningOracle, FLGatekeeper) deployed on *both* a gas-free (PureChain, PoA^2) and a gas-based (Polygon Amoy, PoS) chain—the first side-by-side cost comparison under both economic models; and (4) a **dual-domain evaluation** on a 9-client N-BaIoT [12] autoencoder federation, cross-validated on a 5-client CASAS BiLSTM classifier, across three approximate-unlearning paradigms plus an exact-retraining baseline, measuring MIA forgetting, on-chain cost on both chains, and post-revocation utility recovery.

II. Related Work

Three converging research threads form the prior art for ConsentChain; we summarise each with its gap, then position ConsentChain against all three.

Consent management on distributed ledgers. MedRec [4] records patient consent immutably on-chain—the durability and owner-exclusive control we inherit—but treats the record as the protocol *endpoint*: no training is gated by it, and nothing shows a withdrawal produced a model update (semantic-web vocabularies like GConsent formalise consent off-chain without enforcement). *Gap*: consent is recorded but does not influence training.

Machine unlearning. SISA [5] pre-shards training data so that removal retrains only the affected shard; FedEraser [6]

replays federated aggregation history with a calibration step; and gradient-ascent methods take a few negative-learning-rate steps on the data to be removed. All are validated on standalone benchmarks, but none specifies how unlearning is triggered, by whom, or how a deployer demonstrates it actually happened. *Gap*: unlearning is implementable but not verifiable end-to-end.

Blockchain-based federated learning. BlockFL [7] logs device-side updates on-chain against malicious aggregators and BFLC [8] replaces the aggregator with committee consensus, but neither encodes consent: a joined participant cannot withdraw without trusting the orchestrator. Membership inference [9]–[11] is the canonical test for residual memorisation, yet no blockchain-FL system uses MIA scores as an on-chain acceptance criterion. *Gap*: updates are auditable but consent semantics and forgetting verification are absent.

Positioning. Across the four implicated dimensions—consent representation, unlearning trigger and method, verification, and on-chain footprint—each prior system covers at most one natively. ConsentChain is the first design we are aware of to address all four in a single chain-enforced protocol at gas-free cost: it adopts MedRec's registry pattern (vertex A) under the CLSM safety and liveness properties, accepts any of the three unlearning paradigms as a black box (vertex B), and makes three-MIA verification (vertex C) the chain's gating condition for a consent transition. The resulting loop—consent change $\rightarrow$ unlearning $\rightarrow$ verification $\rightarrow$ attestation $\rightarrow$ finalisation—is the missing edge.

The choice of *which* blockchain matters at IoT scale: permissionless gas-based chains (Ethereum, Polygon) impose recurring per-operation costs that scale linearly with device count and consent operation rate. Section V quantifies this against PureChain [13], a public, gas-free, EVM-compatible chain with PoA^2 consensus and Merkle-proof-backed state. PureChain is not the contribution of this paper; rather, we show the protocol is operationally identical on both chain types while the economic model is fundamentally different.

III. System Model and Threat Model

A. Entities

ConsentChain has six entities. The data subject DS_i controls the private key of one soulbound Consent NFT; the data controller DC orchestrates the FL round loop and triggers unlearning on revocation; each participant device FL_k runs local SGD; the Verifier Committee $VC = \{v_1, \ldots, v_n\}$ holds $n = 5$ registered ECDSA keys with default threshold $t = 3$; and three on-chain contracts—`ConsentRegistry` (CR), `UnlearningOracle` (UO), and `FLGatekeeper` (GK)—collectively implement the protocol. Anyone with chain read access is an *auditor*: public auditability is a first-class property.

B. Federated learning baseline

We adopt FedAvg [1]: with $\theta^{(r)}$ the global parameters at round r, K_r the set of validated participants returned by GK.`validateRound`, and n_k the local training-set size of client k, the update is $\theta^{(r+1)} = \sum_{k \in K_r} (n_k / \sum_{j \in K_r} n_j)\, \theta_k^{(r+1)}$. The novel restriction is that K_r must be the `validateRound` return value rather than an orchestrator-chosen subset: consent is enforced at the K_r membership level, not at the gradient level.

C. Consent Lifecycle State Machine

Definition 1 (CLSM)**.** *The Consent Lifecycle State Machine is the deterministic finite automaton* $\text{CLSM} = (S, \Sigma, \delta, s_0, F)$ *with*

$$S = \{\textsc{Inactive}, \textsc{Active}, \textsc{Suspended}, \textsc{Revoked}, \textsc{Expired}, \textsc{Unlearned}\},$$
$$\Sigma = \{\texttt{grant}, \texttt{modify}, \texttt{suspend}, \texttt{resume}, \texttt{revoke}, \texttt{tick}, \texttt{expire}, \texttt{unlearn_verified}\},$$

$s_0 = \textsc{Inactive}$*, and* $F = \{\textsc{Unlearned}\}$*. Of the nine* δ *transitions,* `grant` *activates a token;* `modify` *and* `tick` *are* Active *self-edges;* `suspend` *and* `resume` *toggle* Suspended*;* `revoke` *and* `expire` *leave* Active *for* Revoked *and* Expired*; and* `unlearn_verified` *(oracle-only) moves either of those to the absorbing terminal state* Unlearned.

`ConsentRegistry` instantiates CLSM exactly: every δ edge is one external function, and the soulbound constraint (no `transferFrom`) ensures only DS_i may transition token i.

D. Threat model

We defend five in-scope adversaries: a *lazy orchestrator* (GK reverts on pending unlearning and emits `ComplianceViolation` after the $\Delta_{\max}$ timeout), a *deceptive controller* (three MIAs plus VC t-of-n signatures), a *consent forger* (soulbound NFT and `dsOwner` check), and a *replay attacker* (UA binding $(\text{chainid}, \text{tokenId}, \text{hashPre})$); colluding participants are out of scope (an orthogonal DP layer). Six further classes are out-of-scope (Section VI); the full per-adversary model is in supplementary `docs/threat-model.md`.

E. Formal properties

ConsentChain must satisfy four properties. **Consent integrity (safety):** every participant admitted to a round by GK.`validateRound` is Active with the round purpose in its consent purpose set, and no δ-path returns Revoked to Active. **Unlearning enforceability (liveness):** within $\Delta_{\max}$ blocks of a revocation, either a valid UA drives the token to Unlearned or the gatekeeper logs a public `ComplianceViolation`—degrading to "no training" rather than "unconsented training." **Unlearning verifiability:** the oracle accepts a UA iff all three MIA scores fall below τ and t-of-n registered verifiers sign it (concrete predicate in Section IV). **Audit completeness:** every consent transition, attestation, `ComplianceViolation`, and per-round model-hash update emits an indexed Solidity event. Safety and audit hold by construction from the contract code; liveness and verifiability are demonstrated empirically in Section V, with full mechanised proofs left to future work.

IV. The ConsentChain Framework

We detail the three layers of ConsentChain: the CLSM and its on-chain instantiation, the Verifiable Unlearning Protocol with its three MIA components, and the three-contract Solidity architecture that closes the automation loop.

A. Consent Lifecycle State Machine

Of the nine δ edges (Definition 1), five are DS_i-controlled (`grant`, `modify`, `suspend`, `resume`, `revoke`), two are time-driven (`tick`, `expire`), and one (`unlearn_verified`) is callable only by the `UnlearningOracle` after a valid UA submission. The safety property follows by inspection: no edge enters ACTIVE from REVOKED or EXPIRED, the `tick`(r) self-edge guards on $\rho_r \in \mathcal{P}$, and `validateRound` (Section IV-C) fires only that edge.

B. Verifiable Unlearning Protocol

On transition to REVOKED or EXPIRED, the gatekeeper sets `pendingUnlearningSince`$[i] = b$ and `validateRound` blocks all rounds (the liveness property). Within $\Delta_{\max}$ blocks the controller must execute unlearning, score the post-unlearning model with the three MIAs, assemble the Unlearning Attestation, gather t-of-n verifier signatures, and submit it.

1) Approximate unlearning methods: ConsentChain is method-agnostic; we evaluate the three families introduced in Section II. SISA-Fed [5] retrains only the shards that touched DS_i's data; gradient ascent applies k negative-learning-rate steps $\theta^{(j+1)} = \theta^{(j)} + \eta \nabla \mathcal{L}(\theta^{(j)}; \mathcal{D}_i)$ ($k = 10$, $\eta = 10^{-4}$); and FedEraser [6] replays the aggregation history excluding DS_i with a 3-round calibration.

2) Three independent membership inference attacks: We evaluate the post-unlearning model f_{post} against DS_i's training set $\mathcal{D}_i$ (the member set M) with three attacks chosen to be statistically independent, so a false negative in one is unlikely to recur in the others; each returns a score in $[0, 1]$, higher meaning more residual membership signal. The *shadow-model* attack [9] is the supervised baseline: $S = 5$ shadow autoencoders trained on auxiliary disjoint data supply "in"/"out" per-feature reconstruction errors that train an MLP attack classifier, and s_{sh} is the fraction of M it flags "in." *LiRA* [10], the strongest attack, calibrates a *per-example* likelihood ratio $\Lambda(x)$ rather than a global threshold: for each $x \in M$ it fits in-/out-pool Gaussians to the log-reconstruction-errors of $n_{\text{in}} = n_{\text{out}} = 4$ reference models, and s_{lira} is the empirical TPR at an α-controlled FPR (canonical $\alpha = 0.05$; $\alpha = 0.5$ ablation in Section VI). The *label-only* attack [11], assuming access only to the reconstruction decision, calibrates a threshold at FPR 0.05 and takes s_{lo} as the fraction of M whose reconstruction error falls below it.

3) Unlearning Attestation and hash binding: The Unlearning Attestation is the tuple $u = \langle i, h_{\text{pre}}, h_{\text{post}}, m, s^{\text{fp}}_{\text{sh}}, s^{\text{fp}}_{\text{lira}}, s^{\text{fp}}_{\text{lo}}, t_u \rangle$: the revoked token ID, SHA-256 hashes of the pre-/post-unlearning model states, the method $m \in \{0, 1, 2\}$, the fixed-point MIA scores $s^{\text{fp}}_j = \lfloor 10^4 s_j \rfloor$, and a timestamp. Every verifier signs the UA hash $h_u = \text{keccak256}(c \,\|\, i \,\|\, h_{\text{pre}} \,\|\, h_{\text{post}} \,\|\, m \,\|\, s^{\text{fp}}_{\text{sh}} \,\|\, s^{\text{fp}}_{\text{lira}} \,\|\, s^{\text{fp}}_{\text{lo}} \,\|\, t_u)$, where $c =$ `block.chainid`; including the chain identifier c binds the attestation to a single chain and prevents cross-chain replay.

The oracle then accepts (u, Σ) under four conditions: all three fixed-point MIA scores are below $\tau = 0.55$ (i.e. $s^{\text{fp}}_j < 5500$); at least t distinct registered verifiers, recovered from the signatures Σ, signed h_u; the pre-hash h_{pre} equals the model hash written once at revocation (CR.`modelHashAtRevocation`(i)); and the token is still REVOKED or EXPIRED. The last two conditions bind the attestation to the exact model version and block re-attestation of an already-UNLEARNED token, closing the replay loop.

C. Three-contract Solidity architecture

ConsentChain is three Solidity 0.8.24 contracts (full NatSpec in supplementary `docs/contracts-spec.md`). `ConsentRegistry` is the soulbound ERC-721 [14] instantiating CLSM, carrying a write-once `modelHashAtRevocation` and a `transitionToUnlearned` callable only by the oracle. `UnlearningOracle` holds VC, t, and $\tau^{\text{fp}} = 5500$; its `submitAttestation` evaluates the four acceptance conditions on-chain and, on success, atomically finalises the token and unblocks the gatekeeper. `FLGatekeeper` gates rounds: `validateRound` reverts if any participant is ineligible or `pendingUnlearningCount>0` (the on-chain encoding of the liveness property), and permissionless `reportTimeout` emits `ComplianceViolation` past the $\Delta_{\max}$ deadline.

a) Automation loop.: A revocation event at the registry fires `recordPendingUnlearning` on the gatekeeper; after the data controller submits a valid UA the oracle reverses this through `unblock`. Registry state thus drives the gatekeeper's behaviour, and the oracle is the only legitimate path to release pending-unlearning blocks—the closed consent→unlearn→verify→attest →finalise loop that is the edge ConsentChain closes.

V. Evaluation

We evaluate ConsentChain on four axes: unlearning effectiveness (E6), on-chain cost on PureChain versus Polygon Amoy (E7), scalability of round validation to $N = 500$ (E8), and utility recovery through revocation; two short on-chain demos additionally validate the liveness and verifiability properties. The full reproducibility recipe is in supplementary `docs/eval-protocol.md`.

A. Experimental setup

The primary federation is a 9-client N-BaIoT autoencoder anomaly detector [12], one client per IoT device; the global model is a symmetric MLP autoencoder ($115 \to 75 \to 50 \to 25 \to 10$) trained with FedAvg over 30 rounds (Flower [15], seed 42). At round 30 we simulate a GDPR Art. 17 erasure against client 5 and trigger one of the three approximate methods or, as the exact reference, full 8-client retraining from scratch.

TABLE I
UNLEARNING EFFECTIVENESS (E6). MIA SCORES AGAINST THE REVOKED CLIENT'S TRAINING DATA, EVALUATED ON THE POST-UNLEARNING MODEL. THE ON-CHAIN GATE ACCEPTS A UA IFF ALL THREE ARE BELOW $\tau = 0.55$.

Method	Shadow MIA	LiRA	Label-only	On-chain accept
No unlearning (baseline)	0.5455	0.0475	0.0060	✓
Gradient ascent	0.4380	0.0435	0.0030	✓
SISA-Fed	0.2542	0.0795	0.0862	✓
FedEraser	0.5232	0.0325	0.0050	✓
Naive retrain (exact ref.)	0.5843	0.0445	0.0138	×

TABLE II
PER-OPERATION GAS AND COST (E7), LIFECYCLE ×10 MEAN. PURECHAIN COSTS ARE EXACTLY \$0; AMOY COSTS ARE NON-ZERO. PER-DS LIFECYCLE = SUM OF THE FOUR OPS.

Operation	PureChain gas	Amoy gas	Amoy MATIC	Amoy USD
`updateModelHash`	30,074	32,673	0.0010	\$0.00009
`grantConsent`	144,868	153,568	0.0046	\$0.00044
`revokeConsent`	107,904	96,404	0.0029	\$0.00028
`submitAttestation` (3-of-5)	102,903	137,600	0.0041	\$0.00039
Per-DS lifecycle	**385,750**	**420,245**	**0.0126**	**\$0.00120**

On-chain experiments run against live contracts on gas-free PureChain (PoA^2) and gas-based Polygon Amoy (PoS), with VC five ECDSA keys, $t = 3$, and $\Delta_{\max} = 1000$ blocks; every run records an `env.json` fingerprint (commit, library versions, hardware, seed).

B. Unlearning effectiveness

Table I reports each method's three independent MIA scores against the revoked client's member set; the on-chain `submitAttestation` predicate accepts a UA iff all three sit below $\tau = 0.55$.

Three observations stand out. All three approximate methods produce UAs the chain accepts (shadow $0.438/0.254/0.523$ for gradient ascent/SISA-Fed/FedEraser; LiRA and label-only well below τ). The hero result is that the naive-retrain baseline—a model that *never saw* client 5's data—yields shadow $0.584 > \tau$ and is *rejected*: shadow picks up client 5's distinctive traffic distribution, not membership, a known confound [9] that LiRA (0.045) and label-only (0.014) do not exhibit; the gate is thus faithful to MIA evidence, not the controller's claim (Section VI). Gradient ascent's shadow even sits *below* naive retrain's—anti-fitting moves the model further from client 5's distribution than never seeing it—tracing a forgetting–utility frontier (SISA-Fed strong-forgetting; gradient ascent/FedEraser preserve-utility). All three UAs executed end-to-end on PureChain, finalising the token to UNLEARNED in the attestation transaction (records in supplementary `onchain_attestation.json`).

C. On-chain cost and scalability

Table II reports per-operation gas usage and cost from a lifecycle ×10 variance run (E7; USD at POL/USD $= 0.095347$, CoinGecko, measurement day).

Per-data-subject lifecycle cost is \$0 on PureChain and \$0.00120 on Amoy. Gas counts differ modestly between the two chains, but the dominant factor at scale is gas *price* (zero versus ~ 30 gwei), not gas count.

`validateRound`(N) scales exactly linearly in the participant count. PureChain runs the full $N \in [5, 500]$ sweep live ($R^2 \approx 1$); Amoy, measured at $N \in \{5, 10\}$ and extrapolated linearly, projects to \$0.0176 per $N = 500$ round. At a typical 100-round/day cadence that is \$1.76/day per deployment on Amoy versus \$0 on PureChain: gas-free infrastructure is a different cost class, not an optimisation. Two further PureChain runs confirm the formal properties empirically: a $\Delta_{\max} = 2$ timeout fires `ComplianceViolation` (liveness), and a t-of-5 signature sweep exercises both branches of the acceptance predicate (verifiability).

D. Utility recovery after revocation

Resuming FL for ten rounds after dropping client 5, the post-revoke model reaches reconstruction error 0.4720, 6.1% *better* than the 9-client baseline (0.5025)—client 5 was an outlier whose removal tightens the fit. Gradient ascent (0.4726) and FedEraser (0.4698) match this at no utility cost, while SISA-Fed pays $+29\%$ (0.6081 by round 40), consistent with its strongest-forgetting position and lowest shadow MIA in Table I.

E. Cross-domain validation (CASAS)

On CASAS (5-client BiLSTM), the distribution-shift confound is sharper still: shadow rejects every method but SISA-Fed ($0.499/0.035/0.000$), the sole gate-passer, consistent with N-BaIoT—full cross-domain results appear in the journal version.

VI. DISCUSSION

Limitations. Shadow MIA conflates a client's distributional uniqueness with membership—naive retrain scored $0.584/0.846$ (N-BaIoT/CASAS) despite never seeing the data—but this is conservative by design: a compliance gate should prefer false rejection to false acceptance, and requiring *all three* MIAs below τ deliberately errs safe. LiRA resists the confound but is α-sensitive, so deployments co-calibrate (α, τ) per model class. ConsentChain accepts *approximate* unlearning—exact retraining is infeasible at IoT scale, and GDPR Art. 17's "reasonable measures" is met by the on-chain acceptance predicate with the chain as audit evidence—leaving formal-erasure guarantees and ECDSA verifier-identity leakage (BLS-collapsed in the journal version) out of scope. Off-chain training on revoked data is unpreventable but leaves no UA binding the deployed model hash to a valid consent state, a publicly detectable omission.

Regulatory & deployment. ConsentChain discharges GDPR Arts. 7/17/21, EU AI Act Arts. 15/26, and CCPA § 1798.105 through the soulbound NFT, the $\Delta_{\max}$-bounded liveness and verifiable-unlearning predicates, and audit completeness (the four properties above; mapping in supplementary `docs/regulatory-alignment.md`). PureChain's PoA^2 is validator-permissioned but publicly readable and submittable (apt for known-stakeholder IoT consortia); the Amoy deployment shows the identical contracts on a permissionless PoS

chain at non-zero cost—the economic model is the decision variable, not the protocol semantics.

Future work: (i) a zk-SNARK over the CLSM transition predicate so `isEligible` proves eligibility without leaking the subject's identity or purposes; (ii) PoA^2-native verifiers re-running the MIAs at block production; and (iii) cross-chain consent portability via a light-client bridge.

VII. Conclusion

We presented ConsentChain, the first protocol we are aware of to bind dynamic consent, machine unlearning, and on-chain verification into a single chain-enforced pipeline, closing the consent–unlearning–verification triangle that prior work addressed only at the vertices. It decomposes into a formally-stated Consent Lifecycle State Machine implemented as soulbound ERC-721 tokens, a method-agnostic Verifiable Unlearning Protocol gated on three independent MIA scores and a t-of-n ECDSA committee signature, and a three-contract Solidity architecture that runs unchanged on gas-free and gas-based EVM substrates. Evaluation on a 9-client N-BaIoT [12] federated autoencoder, cross-validated on a 5-client CASAS BiLSTM activity classifier, confirms four headline numbers: per-data-subject on-chain lifecycle cost of \$0 on PureChain versus \$0.00120 on Polygon Amoy; `validateRound` scaling linearly to $N = 500$ at \$0 on PureChain versus a projected \$0.0176 on Amoy; MIA scores driven below the on-chain $\tau = 0.55$ threshold by approximate unlearning (all three methods on N-BaIoT, SISA-Fed on CASAS); and an absolute utility delta of $\leq 3\%$ on two of three methods—in fact a 6.1% utility *gain* when the revoked client was an outlier. Future work targets BLS threshold-signature aggregation, zero-knowledge consent eligibility proofs, PoA^2-native verifier-committee integration, and cross-chain consent portability.

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